CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet Polymorphic Practice

1. Consider the following classes:

```java
abstract class Pet {
    public void speak() {
        System.out.print("Pet ");
        sound();
    }
    public abstract void sound();
}

class Dog extends Pet {
    public void sound() {
        System.out.print("Woof ");
    }
}

class Cat extends Pet {
    public void sound() {
        System.out.print("Meow ");
    }
}
```

For each line below, indicate Compile/Runtime error or the exact output:

```java
Pet a = new Dog();
a.speak();
```

```java
Pet b = new Cat();
b.speak();
```

```java
Dog c = new Dog();
c.speak();
```

```java
Pet a = new Dog();
a.sound();
```

```java
Pet b = new Cat();
b.sound();
```

2.

```
abstract class Pet {
    public void speak() {
        System.out.print("Pet ");
        sound();
    }
    public abstract void sound();
}

class Dog extends Pet {
    public void sound() {
        System.out.print("Woof ");
    }
}

class Cat extends Pet {
    public void sound() {
        System.out.print("Meow ");
    }
}
```

```
Pet p1 = new Dog();
((Dog) p1).sound();
```

```
Pet p2 = new Dog();
((Cat) p2).sound();
```

```
Dog d3 = new Pet();
```

```
Cat c1 = new Dog();
```